

Exercise 1: Translational Symmetry & Single-Magnon Excitations

Consider a 1D spin- $\frac{1}{2}$ chain of N sites with periodic boundary conditions (PBC), described by the translation operator T :

$$T|s_1, s_2, \dots, s_N\rangle = |s_N, s_1, \dots, s_{N-1}\rangle, \quad \text{where } s_j \in \{\uparrow, \downarrow\}. \quad (1)$$

Questions

- 1.1 Spectrum of T :** Show that $T^N = \mathbb{I}$ and derive the allowable eigenvalues $\lambda_k = e^{-ik}$ of T , expressing k in terms of discrete quantum numbers.
- 1.2 Commutation with H_{XXX} :** For the periodic XXX Hamiltonian $H_{XXX} = J \sum_{j=1}^N \mathbf{S}_j \cdot \mathbf{S}_{j+1}$ (with $\mathbf{S}_{N+1} \equiv \mathbf{S}_1$), prove that $[H_{XXX}, T] = 0$. What is the physical consequence?
- 1.3 Single-Magnon State:** Let $|\mathbf{0}\rangle = |\uparrow\uparrow \dots \uparrow\rangle$ be the ferromagnetic reference state.
- (a) Verify that $H_{XXX}|\mathbf{0}\rangle = E_0|\mathbf{0}\rangle$ and find E_0 .
 - (b) Define single-flip momentum states $|k\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^N e^{ikj}|j\rangle$, where $|j\rangle = S_j^-|\mathbf{0}\rangle$. Show that $T|k\rangle = e^{-ik}|k\rangle$.
 - (c) Derive the dispersion relation $E(k) - E_0$ for a single magnon excitation.

Exercise 2: Global $SU(2)$ Symmetry of the XXX Chain

Define total spin operators $S^\alpha = \sum_{j=1}^N S_j^\alpha$ for $\alpha \in \{x, y, z\}$, and total spin squared $\mathbf{S}^2 = (S^x)^2 + (S^y)^2 + (S^z)^2$.

Questions

- 2.1 Commutation Relations:** Prove that $[H_{XXX}, S^\alpha] = 0$ for all $\alpha \in \{x, y, z\}$, and show that $[H_{XXX}, \mathbf{S}^2] = 0$.
- 2.2 Multiplet Structure:** Explain why energy levels must form $SU(2)$ representations of dimension $(2S+1)$. Prove that if $|\psi\rangle$ is an eigenstate with energy E , then $S^\pm|\psi\rangle$ are also eigenstates with the exact same energy E .
- 2.3 Dimension Reduction:** How does $SU(2)$ symmetry simplify the exact diagonalization of H_{XXX} ?

Exercise 3: Symmetry Reduction in the XXZ Chain

The anisotropic XXZ spin chain Hamiltonian is:

$$H_{XXZ} = J \sum_{j=1}^N \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z \right), \quad \Delta \neq 1. \quad (2)$$

Questions

- 3.1 Continuous Symmetry Reduction:** Prove that for $\Delta \neq 1$, $[H_{XXZ}, S^z] = 0$, but $[H_{XXZ}, S^x] \neq 0$. What is the remaining continuous symmetry group?
- 3.2 Spin-Flip Parity (Discrete Symmetry):** Define $\mathcal{P} = \prod_{j=1}^N \sigma_j^x$, where $\sigma_j^x = 2S_j^x$.
- (a) Show that $\mathcal{P}S_j^z\mathcal{P}^\dagger = -S_j^z$ and $\mathcal{P}S^z\mathcal{P}^\dagger = -S^z$.
 - (b) Prove that $[H_{XXZ}, \mathcal{P}] = 0$.
 - (c) Does $\mathcal{P}$ commute with S^z ? What does this imply for simultaneous quantum numbers?
- 3.3 Spatial Inversion Parity:** Let I be spatial inversion ($IS_j^\alpha I^\dagger = S_{N-j+1}^\alpha$). Show that $[H_{XXZ}, I] = 0$ and derive $ITI^\dagger = T^{-1}$. What does this imply for $E(k)$ versus $E(-k)$?

Exercise 4: The Spin to Hard-Core Boson Mapping

Consider a 1D lattice with spin-1/2 operators $\mathbf{S}_j = \frac{1}{2}\boldsymbol{\sigma}_j$ at each site $j \in \{1, \dots, N\}$. Define the boson creation and annihilation operators $a_j^\dagger, a_j$ via:

$$a_j^\dagger = S_j^+ = S_j^x + iS_j^y, \quad a_j = S_j^- = S_j^x - iS_j^y. \quad (3)$$

Questions

- 1.1 On-site Algebra:** Show that a_j and $a_j^\dagger$ satisfy on-site anti-commutation relations:

$$\{a_j, a_j^\dagger\} = 1, \quad (a_j^\dagger)^2 = 0, \quad (a_j)^2 = 0. \quad (4)$$

What is the physical interpretation of $(a_j^\dagger)^2 = 0$?

- 1.2 Inter-site Algebra:** Show that for different sites ($j \neq k$), a_j and $a_k^\dagger$ commute:

$$[a_j, a_k^\dagger] = 0, \quad [a_j, a_k] = 0 \quad (j \neq k). \quad (5)$$

Explain why these operators are called **hard-core bosons**.

- 1.3 Hamiltonian Mapping:** Map the XXZ Hamiltonian with a transverse magnetic field h :

$$H = -J \sum_{j=1}^{N-1} \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z \right) - h \sum_{j=1}^N S_j^z, \quad (6)$$

into a system of hard-core bosons. Express the parameters of the boson model (hopping amplitude t , nearest-neighbor interaction V , chemical potential μ) in terms of J, Δ , and h .

Exercise 5: The Jordan–Wigner Transformation

Hard-core bosons commute at different sites, whereas true physical fermions must anti-commute across all sites. To map hard-core bosons to spinless fermions $c_j, c_j^\dagger$, we attach a non-local phase factor (the Jordan–Wigner string).

Define:

$$c_j = \exp \left(i\pi \sum_{l=1}^{j-1} n_l \right) a_j, \quad c_j^\dagger = a_j^\dagger \exp \left(-i\pi \sum_{l=1}^{j-1} n_l \right), \quad (7)$$

where $n_l = a_l^\dagger a_l = c_l^\dagger c_l$.

Questions

2.1 Fermionic Anti-commutation: Show that $K_j = \exp\left(i\pi \sum_{l=1}^{j-1} n_l\right) = \prod_{l=1}^{j-1} (-\sigma_l^z)$.

2.2 Prove that the operators $c_j, c_k^\dagger$ satisfy canonical fermionic anti-commutation relations:

$$\{c_j, c_k^\dagger\} = \delta_{jk}, \quad \{c_j, c_k\} = 0. \quad (8)$$

2.3 Local Bilinear Invariance: Prove that nearest-neighbor hopping of hard-core bosons maps directly to local fermionic hopping:

$$a_j^\dagger a_{j+1} = c_j^\dagger c_{j+1}. \quad (9)$$

Explain why the non-local string cancels out for nearest neighbors.

Exercise 6: Exact Solution of the Transverse Field Ising Model (TFIM)

Consider the 1D Transverse Field Ising Model on an open chain:

$$H_{\text{TFIM}} = -J \sum_{j=1}^{N-1} \sigma_j^x \sigma_{j+1}^x - h \sum_{j=1}^N \sigma_j^z. \quad (10)$$

Questions

3.1 Jordan–Wigner Representation: Express σ_j^x and σ_j^z in terms of fermionic operators $c_j, c_j^\dagger$. Show that H_{TFIM} maps to a quadratic fermionic Hamiltonian (Kitaev chain):

$$H_{\text{TFIM}} = -J \sum_{j=1}^{N-1} \left(c_j^\dagger c_{j+1} + c_j^\dagger c_{j+1}^\dagger + \text{h.c.} \right) - 2h \sum_{j=1}^N \left(c_j^\dagger c_j - \frac{1}{2} \right). \quad (11)$$

3.2 Fourier Transform (Periodic Chain): Assuming periodic boundary conditions ($c_{N+1} \equiv c_1$) and introducing momentum operators $c_j = \frac{1}{\sqrt{N}} \sum_k e^{ikj} c_k$, write H_{TFIM} in momentum space as:

$$H_{\text{TFIM}} = \sum_{k>0} \begin{pmatrix} c_k^\dagger & c_{-k} \end{pmatrix} \mathcal{H}(k) \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}, \quad (12)$$

and find the 2×2 matrix $\mathcal{H}(k)$.

3.3 Bogoliubov Spectrum & Quantum Phase Transition:

(a) Diagonalize $\mathcal{H}(k)$ using a Bogoliubov transformation:

$$\eta_k = u_k c_k - i v_k c_{-k}^\dagger, \quad (13)$$

and derive the excitation spectrum $\varepsilon(k) = 2\sqrt{J^2 + h^2 - 2Jh \cos k}$.

(b) Find the critical field h_c where the excitation gap vanishes at $k = 0$. What physical transition occurs at $h = h_c$?